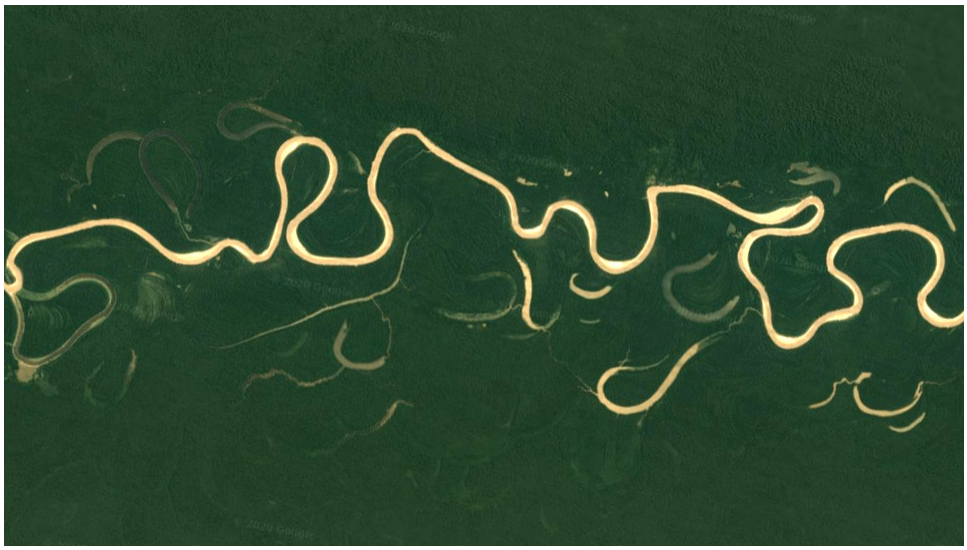
Dynamic Planet C

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Directions:

- There are **97 points** and **5 sections**.
- DO NOT WRITE ON THE EXAM. Please use the answer sheet instead.
- Be clear but concise on explanations–don't write more than you need, but make sure you clearly answer the question.
- Show your work for math and calculation questions unless otherwise stated.

Section I (23 points)

1. (5 points) Match the following drainage patterns to the environments in which they would most likely form: Centripetal, deranged, radial, trellis, dendritic.

    1. Relatively flat tundra with eskers and recently-deposited glacial erratics
    2. Oceanic hotspot volcano
    3. Endorheic desert with salt flats
    4. Region with tightly folded bedrock that has been uplifted to the surface
    5. Mountainous region with granite bedrock

2. Map 1 shows a portion of the San Joaquin River in the central valley of California. Refer to Map 1 for the following questions:

    (a) (1 point) What do the symbols near (a) indicate?

    (b) (3 points) What feature is located near (b)? Briefly explain how the specific feature at (b) might form given the surrounding environment and the climate of central California.

    (c) (1 point) What type of lake might form at (c) if the meander cuts itself off?

    (d) (3 points) (TB) What features are indicated by the straight blue lines near (d)? Briefly explain why they are important to the local community, considering the surrounding environment and the climate of central California.

    (e) (2 points) The dashed line through (e) indicates the boundary of a land tract. Why might it have the shape that it does?

3. (a) (2 points) (TB) The length of the longest stream in a drainage basin is known as the mainstream length, L_m. If the mainstream length depends only on the drainage area A, write a dimensionally-consistent scaling law for L_m in terms of A down to a dimensionless constant k. Briefly explain your answer.

    (b) (2 points) In reality, L_m is empirically related to A by the following relation: $L_m = kA^{0.6}$. What does this indicate about the scaling law you found in (a)? (*Hint*: consider the assumptions made in (a))

4. (a) (2 points) Which of the following can be unambiguously compared by knowing the Strahler stream orders of two streams with *distinct* orders (e.g., 2 and 3) within the same watershed?

        A. Relative flow velocity
        B. Relative discharge
        C. Relative number of tributaries upstream
        D. Absolute number of tributaries upstream

    (b) (2 points) The Amazon River has a Strahler order of 12, the largest in the world! For a stream of Strahler order 12, what is the *minimum* number of index 1 tributaries that must feed it? (You do not need to show your work)

Section II (25 points)

1. (a) (2 points) Which of the following best describes glacial streams?

   A. Braided with peak discharge in late summer

   B. Braided with peak discharge in early winter

   C. Meandering with peak discharge in late summer

   D. Meandering with peak discharge in early winter

   (b) (1 point) Why do glacial streams typically have a milky gray color?

   A. They contain a significant amount of suspended silt

   B. They are saturated with dissolved ions

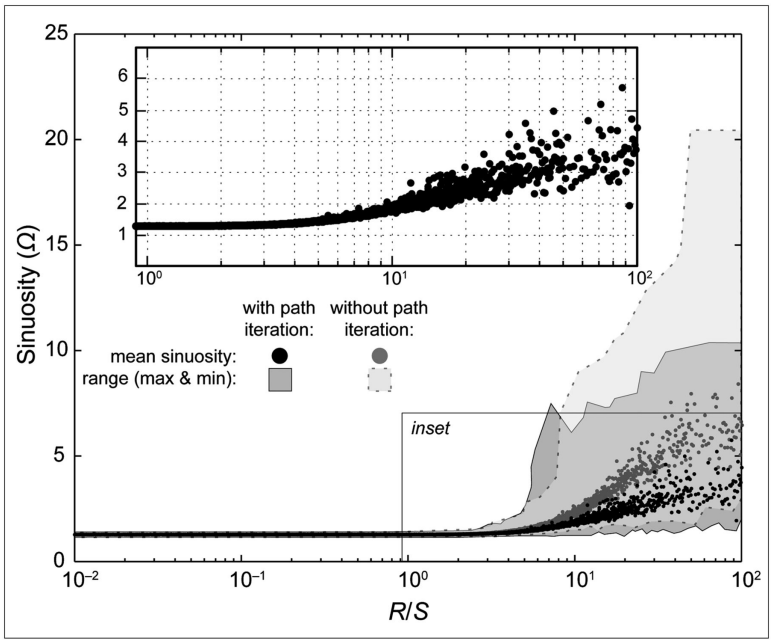
   C. They are saturated with dissolved oxygen

   D. They carry a significant bed load

2. Sinuosity is a characteristic of many surface flows.

   (a) (2 points) In words, briefly describe how the sinuosity between two points of a stream would be calculated.

   (b) A simple model to understand physical controls on channel sinuosity can be constructed by considering the slope of the flow surface and a dimensionless "roughness coefficient".



(Lazarus and Constantine, 2013)

   The figure above shows many realizations of flow-routing simulations where the dimensionless roughness (R) and slope (S) of a surface are varied. The resulting sinuosity Ω is plotted against R/S.

   i. (2 points) Explain qualitatively what occurs for $R \ll S$.

   ii. (2 points) (TB) Explain qualitatively what occurs for $R \gg S$.

   iii. (2 points) You might notice that the figure shows two types of runs: those with path iterations and those without. Without path iteration, it was found that flow paths for large R/S tended to cross

over itself and generate unrealistically large sinuosities. For real meandering rivers, briefly describe the process that limits the sinuosity from going to infinity.

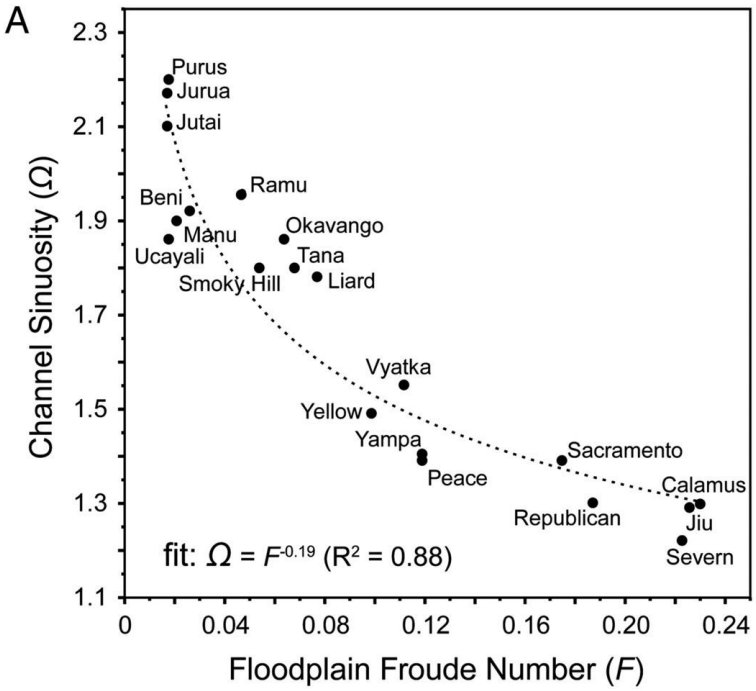
(c) To test the hypothesis that sinuosity is controlled by R/S for real rivers, we need a dimensionless physical parameter analogous to R/S. Fortunately, it exists! For open channel flows, the Froude number is a dimensionless ratio that depends on the slope, roughness, and hydraulic radius of the flow. It is defined as:

$$\mathrm{Fr} = \frac{v}{\sqrt{gR}}$$

where v is the average flow velocity, g is gravitational acceleration, and R is the hydraulic radius.

i. (3 points) Use Manning's equation to derive an expression for the Froude number in terms of slope S, roughness n, hydraulic radius R, and any other physical constants.

ii. (3 points) It turns out that the Froude number is well correlated with the channel sinuosity for many floodplains worldwide:



(Lazarus and Constantine, 2013)

Is the relationship shown above consistent with the R/S-sinuosity relationship established using the flow-routing simulations? Use your result from (i) to justify your answer.

3. (2 points) Which of the following are characteristics of graded streams that drain into the ocean?

A. Contains nick points

B. Passes through several local base levels

C. Both A and B

D. None

4. Refer to Figure 1 for this question.

   (a) (2 points) The image shows the Horton River draining into the Arctic Ocean. Briefly describe how this could have formed.

   (b) (2 points) From satellite observation, the Horton River delta was found to have shrunk in the past decades. Which of the following can be unambiguously concluded from this observation?

      A. The river stopped depositing sediment

      B. The river was rerouted to another path

      C. The rate of erosion by the ocean exceeded the rate of deposition

      D. None of the above

   (c) (2 points) Which of the following are generally true about river deltas?

      1. Deltas form because the competence of the flow decreases suddenly

      2. Deltas typically exhibit a vertically graded structure from multiple episodes of deposition

         A. 1 only

         B. 2 only

         C. 1 and 2

         D. None

Section III (12 points)

1. Consider a flood that has a recurrence interval of 100 years (i.e., it is a 100-year flood).

   (a) (1 point) What is the probability that a 100-year flood is exceeded by another flood in any given year?

   (b) The Poisson distribution is a probability distribution that gives the probability that a number of events occur over some period of time if these events occur at a known constant rate. It is a function of r, the number of occurrences of the event:

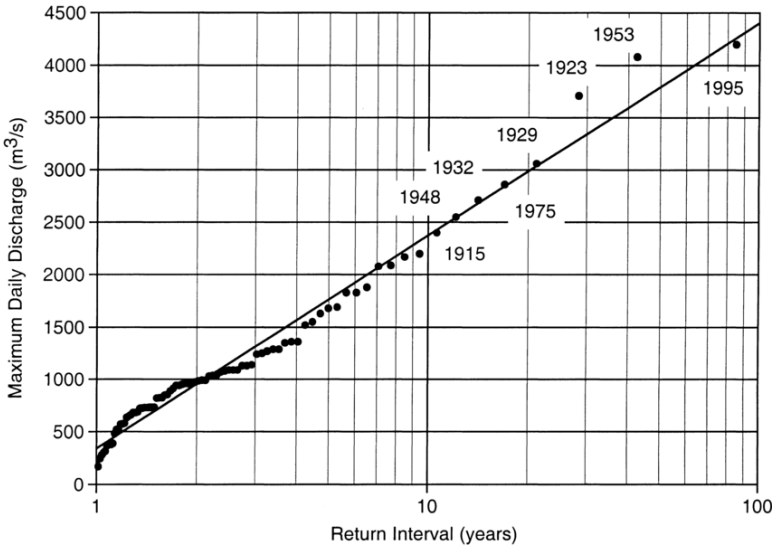
$$P(r) = \frac{(t/T)^r}{r!} e^{-t/T}$$

   where t is the time period of interest and T is the recurrence interval. For these questions, assume that floods follow a Poisson distribution.

      i. (2 points) Calculate the probability that a 100-year flood occurs only once during 1 year.

      ii. (3 points) (TB) Calculate the probability that a 100-year flood occurs at least once in 1 year. (*Hint*: occurrence means $r \neq 0$).

      iii. (2 points) Calculate the probability that a 100-year flood occurs at least once in 100 years.



(Kalischuk et al., 2001)

2. The plot above shows the relationship between the recurrence interval of floods and their maximum discharge for the South Saskatchewan River.

   (a) (2 points) If the maximum discharge is Q and recurrence interval is T, what is the empirical relationship between Q and T?

   A. Q scales linearly with T

   B. Q scales exponentially with T

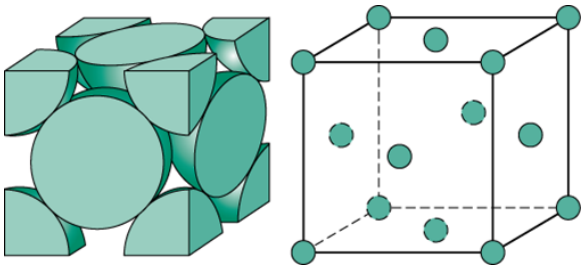
   C. Q scales inversely with T

   D. Q scales logarithmically with T

   (b) (2 points) The river stage is defined as the "height" of its water above some predefined reference. Why might stage not necessarily be a good indicator for understanding flood extent?

Section IV (20 points)

1. (a) (3 points) The porosity ϕ is defined as the void volume V_{void} over the total volume V. Let's consider a well sorted sandstone, so well sorted that the grains are all the same size! They fall into a close packing arrangement (a "face-centered cubic" arrangement) shown below. Calculate ϕ for this sandstone.



   (b) (2 points) Use the model above to justify whether a poorly sorted sandstone would have higher or lower porosity than the well sorted sandstone.

(c) (2 points) Which of the following rocks typically has high porosity but low permeability?

A. Granite

B. Vesicular basalt

C. Sandstone

D. Shale

2. (2 points) Which of the following is equivalent to the water table in an unconfined aquifer?

A. Potentiometric surface

B. Cone of depression

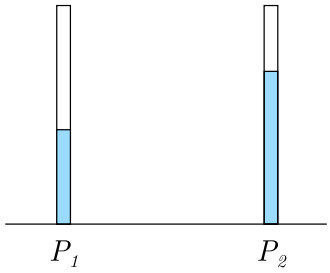
C. Gravimetric surface

D. Artesian surface

3. Hydraulic head is a way to represent the total potential energy of the fluid. It is the height above some predefined reference that a fluid would rise if all of its energy were converted to gravitational potential.

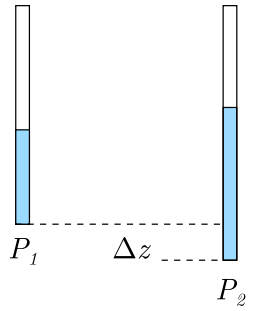
(a) (2 points) For a fluid with a potential energy per unit volume of E, find an expression for the hydraulic head h in terms of E, density ρ, and any physical constants.

(b) (3 points) Consider two piezometers that measure the elevation of groundwater above the same reference height (you can think of this as the base of the aquifer), shown below:



They measure a pressure difference at the base $\Delta P = P_2 - P_1$. Find the difference in hydraulic head Δh between the two piezometers.

(c) (2 points) (TB) It's not very realistic for the piezometers bases to be at the same height, so we'll consider a more general case where their bases are separated vertically by a distance Δz. Find Δh, accounting for the shift Δz.



4. Refer to Figure 2 for the following questions.

   (a) (2 points) Which of the following locations would have the greatest rate of groundwater flow, assuming the hydraulic conductivity is about the same everywhere in this aquifer?

   A. A

   B. B

   C. C

   D. D

   (b) (2 points) The border between Lafayette and Suwannee County is a river. Justify this using observations of the potentiometric contour.

Section V (17 points)

1. (4 points) Lakes can be categorized based on their chemistry and biological productivity. Assign the following lake classifications to the descriptions below: eutrophic, oligotrophic, argillotrophic, dystrophic.

   1. Low biological productivity, acidic, and rich in organic matter
   2. High biological productivity, with abundant N and P
   3. Low biological productivity, with an abundance in clays
   4. Low biological productivity, cold, and clear

2. In this question, you will construct a water budget for a hypothetical lake.

   (a) (3 points) The terms of the water budget are defined: P, the precipitation rate (m/yr); E, the evaporation rate (m/yr); A, the surface area of the lake (m^2); G, the groundwater inflow rate (m^3/yr); R_{in}, the river inflow rate (m^3/yr); and R_{out}, the river outflow rate (m^3/yr).

   Let Q be the change in volume of the lake over time in m^3/yr. Write an expression for Q using the terms defined above.

   (b) (2 points) Some lakes have dried up due to human intervention (diversion of rivers for irrigation, groundwater withdrawal, etc.). For these lakes, the rate of evaporation E typically *increases* as their size decreases. Give a reason for why this might be the case.

3. (3 points) Define a seiche and give two possible ways by which they can be generated.

4. (a) (1 point) Which of the following lake types do not experience mixing?

   A. Holomictic

   B. Meromictic

   C. Monomictic

   D. Dimictic

   (b) (4 points) Consider a typical mid-latitude lake in Wisconsin that freezes over every winter. Describe the seasonal mixing pattern (how many times does it mix per year?) and why mixing occurs.